

CLEAR — Clean, Leakage-free Evaluation And Reporting

an honest, leakage-free evaluation harness for highway trajectory prediction

1 · INPUTS

Dataset (highD / exiD / NGSIM) — you provide, CLEAR ships none

Your model:
`predict(ws) -> (N,T,2)`

2 · CLEAR PIPELINE

- canonical per-window frame (handles curved ramps)
 - leakage-safe split: recording / vehicle-disjoint
-> PASS-CLEAN or WARN-LEAKY
- honest baselines CV / CA / IDM
 - metrics: FDE/RMSE, CIs, CA win-rate, strata, best-of-k, predicted-collision

3 · EVAL CARD

Protocol + leakage verdict
Accuracy ladder vs CA
Stratified error (cruise/hard)
Admissibility (collisions)
Best-of-k transparency

5-point disclosure checklist
JSON · Markdown · figure

WHY

- **Leakage inflates.**
- **CA is a hard baseline.**
- **Averaging hides the tail.**
- **Accuracy != safety.**

Sliding windows share vehicles/recordings across folds; CLEAR forbids it by default.

A zero-parameter constant-acceleration model is unbeaten on clean highway data.

~77% of windows are trivial cruising; one RMSE buries the safety-critical cases.

The lowest-error model can forecast the most rear-end collisions; CLEAR measures both.